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**CUBESAT PAYLOAD: LOW-COST EXPERIMENTAL DETERMINATION OF THE THERMAL
EMISSION OF E-PAPER FOR USE IN LOW POWER SATELLITE ACTIVE TEMPERATURE
REGULATION**

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Commented [MR1]: Is this an accurate title for what you are actually presenting? I'd argue that something like "Experimentally determining the thermal emissivity of e-ink for use in low-power satellite temperature regulation" (or something like that...you can definitely make something more refined than that!) is more representative of what you actually did.

ABSTRACT

In the present study, a low cost means of determining emissivity of some material was developed. It was used to determine the emissivity of e-paper at varying pixel densities shown on screen. The module had significant variance between tests for e-paper.

Keywords: Radiation, Heat Transfer, Small Satellites, E-Paper, Thermal Emissivity, Low-Cost

NOMENCLATURE

k	Thermal Conductivity
LHCM	Linear Heat Conduction Module
D,d	Diameter
ϵ	Emissivity
σ	Stefan-Boltzmann constant
q_z	Axial Heat Flow
z	Axial Distance
T	Temperature
ΔT	Temperature Change
R	Thermal Resistance
C	Thermal Capacitance
LEO	Low Earth Orbit

1. INTRODUCTION

Satellites in low earth orbit can go from an extremely cold environment (-65°C) to an extremely hot one (125°C) within the course of 4 hours as they orbit the Earth [1]. Therefore, satellites are fitted with thermal control systems that remove excess heat during the hot part of the cycle and that retain heat during the cold part of the cycle to keep satellite components, particularly sensitive electronics such as clocks and optical sensors, at a functional temperature. [2,3]

Low power small satellites typically use active thermal control through resistive heaters [4]. Resistive heaters pass current through a conductive metal with high electric resistance to convert this electricity to heat. This, and various other forms of active thermal regulation systems can consume large amounts of power per unit area (intensity), ranging from 0.4 up to 7.75 W/cm^2 [5,6].

This research aims to explore the effectiveness of e-paper as a potential thermal controller for small satellites by developing a low cost means of experimentally determining emissivity of some material. E-paper is an electronic display that consists of a film filled with a clear fluid and charged ink particles. When a charge is applied to the film, it will attract or repel the charged particles, accordingly, raising or repelling the pigment to the surface of the film where it remains until another charge is applied. When the pattern shown on the screen is not being changed, it draws no current [7,8]. This is especially significant because power generation and space is limited on the 1,000 cubic centimeter volume of 1U cubesats.

The necessity for a system that is able to both produce and reject heat is extremely useful for the purpose of cubesats, as when more power is used onboard, more heat is generated. This may contribute to the failure of onboard components as well, for example a component may undergo thermal expansion [9]. Active thermal control that can be achieved without resistive heating allows for the power budget of the cubesat to be redistributed, minimizing power budget on the thermal regulation system, and optimizing this power budget for other subsystems of the satellite.

This research determines the thermal characteristics of e-paper, a step towards the implementation of an active low-power option for satellite thermal regulation. To do this, we developed a thermal model using energy balances. This informed an experimental system consisting of a metal plate and an e-paper module at steady state, in which the thermal impact of radiation on the e-paper display can be quantified. The results define a function that maps the pixel density shown on the screen to the effective coefficient of emissivity.

This lays the foundation for autonomous variable thermal control in small satellites. E- paper can potentially provide alternating heating and cooling with an extremely small power draw, only during transition stages, thereby improving the battery economy of small satellites.

2. MATERIALS AND METHODS

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A benchtop experiment was developed to test the emissive abilities of e-paper. This experiment would be replicated in a benchtop space-like environment, within a vacuum and with a ultraviolet infrared light source to mimic the sun, and then in space. The change in the emissivity of e-paper can be determined through replicating this experiment with varying pixel densities. Figure 1 shows key elements of the benchtop test system.

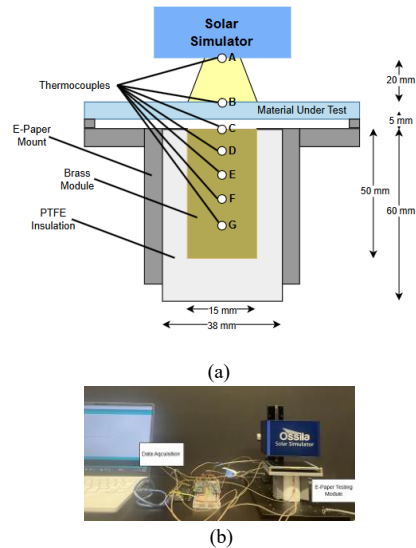


FIGURE 1: (a) BENCHTOP TEST SETUP SCHEMATIC (b) TEST SETUP

The system rests 20mm below the bulb of the solar simulator, to promote the rate of heat exchange while remaining in the constraints of the solar simulator's effective beam diameter.

The light is turned on and remains untouched until steady state is reached. Steady state is determined by calculating the thermal conductivity of brass using the internal temperature measurements and Fourier's law of conductivity and comparing that to the literature value for that of brass.

2.1 Apparatus and Instrumentation

The system is designed to replicate a Linear Heat Conduction Module (LHCM). The LHCM presents a method to experimentally determine heat flow within an open system to characterizes the thermal conductivity of some material [10,11]. LHCMs consist of several metals of known conductivity, stacked together with the material of interest inserted between the stacks. A heater is inserted into the stack at one end, and a cooling loop into the other, creating a temperature gradient and promoting heat flow throughout the stack.

This project utilized type T thermocouples, which can measure temperatures from -270 C to 370 C within an error of ± 1.1 C or $\pm 0.75\%$ (whichever is greater) [12]. Type T thermocouples were chosen for this experiment specifically because their thermal range of operation covers the range of temperatures experienced in low-earth orbit and accordingly are space compatible. Furthermore, type T thermocouples have the lowest associated error among commercially available thermocouples. A heat shrink layer was added around any exposed thermocouple wire, excluding the junction, to minimize any extraneous contact points between the thermocouple and the brass cylinder.

MCP 9600 Thermocouple Amplifier Chips were used to amplify the voltage drops between the two thermocouple channels [13]. Figure 2 shows the circuit used for data collection.

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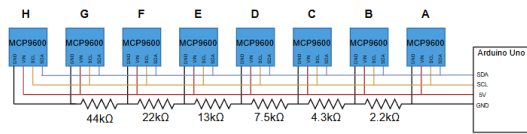


FIGURE 2: DATA ACQUISITION CIRCUIT FOR THERMOCOUPLE AMPLIFIERS

While various e-paper modules should be tested in the future, a waveshare 4.2", 400 x 300 pixel screen in gray scale was used for the present study [14]. This is because of its ease of interfacing with an Arduino and size match to a 1U cubesat. It can be refreshed 1,000,000 times. The module was selected as a proof of concept and works within a temperature only a portion of the range of temperature in LEO, from -25 °C to 70 °C.

An Oscilla LED Solar Simulator was used to mimic sunlight in a controlled environment, and was the main source of heat in the system. It is AAA rated at a 15mm diameter 8.5cm away from the beam, meaning it is a near perfect spectral match to the sun [15]. The usage of solar simulators was cost prohibitive for the present study and the Oscilla Solor Simulator had comparable spectral matches to other solar simulators on the market, but at a fraction of the cost. This solar simulator costs \$2,000, while a Xenon Arc lamp, which may allow for more accurate results, costs on average upwards of \$8,500 considering the cost of both the bulb and the housing. [16]

The brass module is 50mm long and 15mm in diameter to match the effective beam diameter of the solar simulator. Holes with a 3mm diameter were added every 10mm along the length of the rod to insert the thermocouples into the module. These holes were 3mm in diameter to allow for a press fit clearance between the thermocouples and the modules. This means that they can be fastened using only the friction between the sheathing on the thermocouples and the metal. The holes were all 7.5 mm in depth to allow for all temperature readings to be taken at the same point along the central axis of the module. Finally, the 10mm spacing between thermocouples along the length of the rod allowed for various temperature readings to be taken within the brass module.

The e-paper module was fastened to the brass module with a 3D printed mount, and 6mm long brass standoffs.

2.2 Thermal Model and Analysis

In order to understand the dynamics of heat flow within the system, it was simplified using the thermal circuit shown in Figure 3, where R indicates a thermal resistance, T indicates a temperature, Q indicates a heat flow, and C indicates a thermal capacitance.

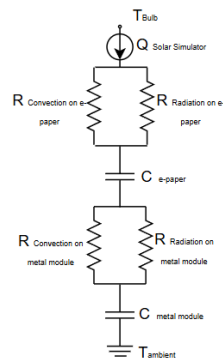


FIGURE 3: CIRCUIT MODEL OF THERMAL SYSTEM

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The system may be treated as a closed system, as mass does not pass between the boundary layer, when the control volume is defined as the volume enclosed by the area directly surrounding the modules and solar simulator. The solar simulator is the main source of heat flow. There is convective and radiative heat transfer occurring between the solar simulator and the e-paper, and convective and radiative heat transfer occurring between the e-paper and the brass module.

With LHCMs, the system is analyzed when steady state is reached and the net heat flow is zero. At this point, the thermal conductivity of the stack, including the unknown material, can be calculated using Fourier's law of conductivity, given in equation 1.

Fourier's law of conductivity, in a single direction, is given by:

$$q_z = -kA_z \frac{\Delta T_z}{\Delta z} \quad (1)$$

where q_z is heat flow along the centerline of the brass module, k is representative of the conductivity of the material, A is representative of the cross-sectional area of the stack, and Δz is representative of the axial distance between the points at which the temperature differential is taken. The brass cylinder works the same way an LHCM does, therefore the heat flow within the cylinder may be determined with this method.

The heat flow due to convection between the e-paper module and the brass cylinder can be estimated using Newton's Law of Cooling, given in equation 2, where the coefficient of convection is estimated to be about 5 W/m²K, indicating free convection of air, the effective area is that of the effective beam of the solar simulator, and the temperature change is that of the surface of the e-paper and the surface of the brass cylinder [17].

$$q_{convection} = h_{free\ convection} A_{effective} \Delta T_{convection} \quad (2)$$

This term is negligible in space, where convection is not a primary mode of heat transfer, and is merely an estimate for that of the benchtop test, as the coefficient of convection used is an estimate and cannot be determined with this experimental setup.

A combination of Fourier's law of conduction and Stefan-Boltzmann's law of radiation can be used to solve for the emissivity of the e-paper at the pixel density being analyzed.

The Stefan-Boltzmann's law of radiation is given by

$$q = \sigma \epsilon A_{surface} T_{surface}^4 \quad (3)$$

where the Stefan-Boltzmann constant is 5.670 374 419... x 10⁻⁸ W m⁻² K⁻⁴, ϵ is the emissivity of the surface, $A_{surface}$ represents the surface area of the body that is emitting radiation, and $T_{surface}$ is the average surface temperature of that body.

At steady state in a closed system, the heat flow due to radiation on the e-paper is equal to the sum of the heat flow due to convection between the e-paper and the brass as well as the heat flow due to conduction in the brass cylinder. Equation 4 gives the emissivity of the e-paper with regards to this relation.

$$\frac{1}{\sigma A_{surface} T_{surface}^4} \left(\frac{-k A_z \Delta T_z}{\Delta z} + h_{free\ convection} A_{effective} \Delta T_{convection} \right) = \epsilon \quad (4)$$

When the system is in a vacuum, however, the effects of convection need not be considered.

For this experiment, the conductive module is made out of 360 Brass alloy, which has a known thermal conductivity of 115 W/mK [15]. Because only a single brass rod was used for the conductivity portion of the system, there are no losses due to contact resistance.

Moreover, because the heat source is only AAA rated for a 15 mm diameter within 85mm of the bulb, rather than taking the entire area of the e-paper, the area of the source was taken to be the effective area of the source. This area was equivalent to that of the brass module, as well as the effective area of convection, simplifying the equation to equation 6.

$$\frac{1}{\sigma T_{surface}^4 \Delta z} (-k \Delta T_z + h_{free\ convection} \Delta T_{convection} \Delta z) = \epsilon \quad (6)$$

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2.3 Experimental Procedure

Before the experiment, all thermocouples must be calibrated. This was done by dipping each thermocouple into an ice bath at 0°C.

Then, all thermocouples were fastened to the module by either inserting them into the holes in the system, or taping them to indicated spots within the system with thermal tape.

Thermocouple A was taped to the surface of the Oscilla Solar Simulator bulb, to experimentally gather the bulb temperature. Thermocouple B was taped to the surface of the e-paper and thermocouple C was taped to the surface of the brass module. Then thermocouples D, E, F, and G were inserted into the holes of the mount in order from top to bottom.

The e-paper module was fastened to the mount, then set to the pattern being tested. To gather control information, only a blank screen and fully colored screen were tested.

The module was then placed in the center of the solar simulator and the height of the bulb with respect to the e-paper was adjusted such that the bulb was no more than 80mm away from the surface of the copper. This ensures that the module experiences the maximum diameter of the AAA waves from the simulator.

Every 10 minutes, the heat flow was determined over several points of the system. When the heat flow within the brass cylinder was equal over the same distances within the cylinder and unchanging, the experiment was concluded, and the final temperatures were noted.

3. RESULTS AND DISCUSSION

Preliminary tests were conducted to ensure that the thermal conductivity of the brass could be experimentally determined with this setup, as this is an input Fourier's law and therefore critical to the analysis of the system.

This entailed collecting data only for the brass module, without the addition of an e-paper module. The experiment was conducted 3 times for the system with brass only and the system with both brass and PTFE insulating layer. Table 1 shows the temperature gradient within the cylinder for either case.

TABLE 1: AVERAGE TEMPERATURE GRADIENT OF PRELIMINARY TESTS AT STEADY STATE

Test Setup	T _D	T _E	T _F
Not insulated	29.41	28.98	28.06
Insulated	39.64	38.92	38.56

The flow between thermocouples D and F are determined using the known thermal conductivity. This is inputted into Fourier's law again, this time to estimate the thermal conductivity over points C and D. Table 2 shows the determined thermal conductivity associated with either case.

TABLE 2: PRELIMINARY TEST RESULTS FOR THERMAL CONDUCTIVITY OF THE BRASS CYLINDER

Test Setup	Experimental conductivity (W/mK)	Actual conductivity (W/mK)	σ	% error

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Commented [MR25]: This is part of your calibration sequence, it doesn't need to be repeated here

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Not insulated	29.82	115	± 5.31	74.07
Insulated	124.25		± 10.11	8.04

The system without insulation was able to determine the thermal conductivity of brass within 74% error at steady state (appx 1.5 hours). However, the experiment was much more effective with an insulation layer, giving a thermal conductivity within 8% error at steady state (appx 1.25 hours).

TABLE 3: RESULTS SHOWING EMISSIVITY ASSOCIATED WITH FULLY COLORED AND FULLY UNCOLORED E-PAPER SCREENS

Test Setup	Experimental emissivity	Standard deviation
Fully uncolored e-paper	0.0102	± 0.00243
Fully colored e-paper	0.0162	± 0.133

The system with a fully uncolored e-paper module was experimentally about equally as emissive as the system with a fully colored e-paper.

However, the ability of the fully colored e-paper to absorb heat was notable during the tests, as the average surface temperature of the uncolored and fully colored e-paper were 44.13°C and 77.84 °C, respectively.

There was significant variance between tests for the fully colored e-paper module. Moreover, there was significant variance in the preliminary thermal conductivity calculation for these cases. This is likely due to the $\pm 1.1^\circ\text{C}$ error associated with the thermocouples, and the fact that the temperature differential was less than this error.

In running the test several times, many points of potential error have been identified. To help verify the accuracy of the thermocouples, an IR camera should be implemented into the procedure, noting the surface temperatures of the tested material and the brass module temperatures at various intervals of time throughout the experiment.

Also, the experiment was intended to be ran in a vacuum in order to treat the effects due to convection as negligible. However, for initial proof of concept purposes, the experiment was conducted outside of a vacuum. Throughout this experiment, the convection between the module and ambient were considered negligible due to the insulation of the brass. The convection between the e-paper module and the brass module was also considered negligible due to the comparatively small space between the two modules.

However, to more accurately characterize the radiatively thermal properties of e-paper, it is critical that other modes of heat transfer are minimized. The results presented in the paper are given under the assumption that there were minimal effects of convection, which is not accurate for LEO.

It may also be beneficial to add a cooling source to the bottom of the brass and PTFE module to help to establish a greater temperature gradient. Currently, the gradient is between the ambient temperature and the temperature of the solar simulator's bulb, from about 50°C to 20°C. The temperature gradient between points may also be less than the error associated with the thermocouples. This compromises the analysis significantly and adding a cooling source may help increase the difference in temperature over the same distances.

In the future, it may also be beneficial to collect temperature points at the connection points between the e-paper and the brass module. Currently, steady state is decided to be the point at which the heat flow within the brass module is equal at all points. Including the heat flow due to conduction between the standoffs and the brass module will allow for a means of connecting the systems in the analysis. They can be used as another data point to verify whether or not the system has reached steady state.

Also, while this system utilized type T thermocouples to better mitigate error, type K thermocouples are more commonly used in space applications. This compromises the accuracy of the tests but will allow for them to be replicated in low earth orbit. [18]

Moreover, as the system is intended to be tested in space, the module must be repackaged accordingly. This includes fastening the thermocouples permanently to the setup, to prevent them from dislodging during launch. This also includes changing the circuit board to a PCB, to prevent any wires from becoming dislodged during launch.

4. CONCLUSION

The present study presented a low cost method of determining the emissivity of some material. This was done for e-paper on a benchtop, to be replicated in LEO on a cubesat. The benchtop experiment produced emissivity coefficients that may not be significant enough to provide active thermal control to a small satellite. However, provided various changes to the setup, this may be a cost effective means of determining emissivity, and in doing so, allow for a low energy active thermal controller for small satellites to be developed using e-paper.

ACKNOWLEDGEMENTS

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